

Methionine or Not Methionine at the Beginning of a Protein

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Summary

*Methionine aminopeptidases with a universal specificity have been revealed from the sequences of the amino-terminal region of mutant forms of yeast iso-1-cytochrome *c* and from a systematic examination of the literature for amino-terminal sequences formed at initiation sites. The aminopeptidase removes amino-terminal residues of methionine when they precede certain amino acids, with a specificity that appears to be determined mainly by the residue adjacent to the methionine residue at the amino terminus. The result with the mutationally altered iso-1-cytochromes *c* and the results from published sequences of other proteins from a wide range of prokaryotes and eukaryotes suggest that the aminopeptidase usually cleaves amino-terminal methionine when it precedes residues of alanine, cysteine, glycine, proline, serine, threonine and valine but not when it precedes residues of arginine, asparagine, aspartic acid, glutamine, glutamic acid, isoleucine, leucine, lysine or methionine. We suggest that the specificity is almost always determined simply by the size of the side chain of the penultimate residue; methionine is usually cleaved from residues with a side chain having a radius of gyration of 1.29 Å or less, but is not cleaved from residues with larger side chains.*

Introduction

The biosynthesis of all proteins from all living cells begins with methionine. In prokaryotes, mitochondria and chloroplasts, translation of proteins is initiated with N-formylmethionine; whereas in the cytosol of eukaryotes, proteins are initiated with just methionine. During and after translation, the amino-terminal regions of most proteins are subjected to a variety of modification and processing events. The N-formyl group is removed from proteins in prokaryotes and organelles by a deformylase enzyme, leaving a methionine residue at amino termini. The amino-terminal methionine can subsequently be cleaved from prokaryotic as well as eukaryotic proteins by a general methionine amino-

peptidase. Further amino-terminal maturation can take place if the protein is secreted, usually by removal of the signal sequence, which is generally 15 to 30 residues in length. In addition, many proteins also are processed by proteolytic cleavage at one or more sites in the polypeptide chain, resulting in the formation of shorter peptides from a single gene product. In addition to proteolytic processing, amino termini can be modified by the covalent attachment of residues and blocking groups. Most notably, amino termini of numerous proteins are acetylated by the action of transacetylases on nascent chains still attached to ribosomes but after other earlier processing of amino-terminal groups. Thus, mature proteins seldom have methionine residues at their amino termini even though they are synthesized that way. In order to help understand amino-terminal processing of proteins, we have been investigating the specificity of methionine aminopeptidase, one of the key enzymes in the processing pathway.

Early work with cell-free systems revealed enzymatic activities that remove amino-terminal residues of methionine from certain proteins in both prokaryotes¹⁻⁴ and eukaryotes.⁵⁻⁷ This excision is believed to occur before the completion of nascent polypeptide chains⁸⁻¹⁰ and before the occurrence of other amino-terminal processing events such as amino-terminal acetylation.^{11, 12} We have used mutant forms of yeast iso-1-cytochrome *c* to investigate the specificity of this methionine aminopeptidase.

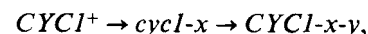
Cytochrome *c* Resembles Cytosolic Proteins

Although eukaryotic cytochromes *c* are a component of mitochondria, they are encoded by nuclear genes and are translated on free cytoplasmic ribosomes. In fact, the mitochondria of ρ^0 mutants that completely lack DNA and are unable to carry out protein synthesis nevertheless contain cytochrome *c* in normal amounts and with normal sequences.¹³ Unlike the import of certain other nuclear-encoded components of

mitochondria, cytochrome *c* import into mitochondria is independent of its amino-terminal region (see below) and does not involve proteolytic processing. The current hypothesis is that transport into the intermembrane space of mitochondria first requires binding of completely translated apo-cytochrome *c* to a mitochondrial receptor and subsequently covalent attachment of the heme group.¹⁴ Thus in some respects cytochrome *c* can be considered a cytosolic protein which responds typically to the action of methionine aminopeptidase.

Altered Iso-1-cytochromes *c*

Bakers' yeast, *Saccharomyces cerevisiae*, contains two forms of cytochrome *c*, iso-1-cytochrome *c* and iso-2-cytochrome *c*, that are encoded respectively by the chromosomal genes *CYC1* and *CYC7*, and that constitute, respectively, 95% and 5% of the total cytochrome *c* complement. These genes and their gene products have been favorites of molecular biologists and geneticists. *CYC1* mutations were defined long before the era of recombinant DNA, primarily because of the ease of isolating and sequencing cytochrome *c* and the ability to deduce nucleotide changes from amino acid changes¹⁵. One major approach was the analysis of the series of mutations of the type



where *CYC1*⁺ denotes the wild-type gene that determines iso-1-cytochrome *c*, *cyc1-x* denotes mutations that cause deficiency or nonfunction of iso-1-cytochrome *c*, and *CYC1-x-y* denotes intragenic reversions that restore at least partial activity and that give rise to either the normal or altered iso-1-cytochrome *c*. Over 500 *cyc1-x* mutants have been isolated on a chlorolactate medium by a method that is based on the finding that mutants partially deficient in cytochrome *c*, containing approximately 5% of the total normal level, are defective in the utilization of lactate but are still able to utilize other nonfermentable substrates such as glycerol or ethanol.¹⁵ These partially deficient mutants unable to utilize lactate

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are resistant to the toxic action of the analogue chlorolactate. Thus chlorolactate medium, which contains chlorolactate and a nonfermentable carbon source such as glycerol or ethanol can be used to enrich for mutants partially defective in cytochrome *c*. As expected, *cycl* mutants which lack iso-1-cytochrome *c* but which retain the normal low-level amount of iso-2-cytochrome *c*, are the major class of cytochrome *c* deficient mutants arising on chlorolactate medium. On the other hand, *CYCI*-*x-y* revertants having iso-1-cytochrome *c* can be selected from the *cycl*-*x* deficient mutants by growth on lactate medium. During the past 20 years over 100 different iso-1-cytochrome *c* sequences have been obtained from *CYCI*-*x-y* revertants. Some of these contain alterations in the amino-terminal region and have been the subject of a study on amino-terminal processing *in vivo*.¹⁸

The sequence of normal iso-1-cytochrome *c* and the sequence for the corresponding mRNA are presented in Fig. 1. Finding threonine at the amino terminus of the mature protein and finding an AUG codon to the left of the threonine codon in the mRNA suggested that a methionine residue was cleaved from the precursor form of the protein.¹⁷ Using the same approach, further deductions on the cleavage of amino-terminal residues of methionine were possible after examining altered forms of iso-1-cytochrome *c*. In fact, functional proteins with any amino terminus probably can be generated because the amino-terminal end of the protein is dispensable for function. A total of 59 different iso-1-cytochromes *c* having replacements, deletions and insertions throughout residue positions 1 to 10 have been shown to be at least partially functional (D. M. Hampsey and F. Sherman, unpublished results).^{18,19} In addition, mutations produced *in vitro* and tested *in vivo* demonstrated that the 5' end of the mRNA encompassing at least nucleo-

tides -32 to +3 can be deleted without appreciably affecting transcription or translation (S. B. Baim and F. Sherman, unpublished results). In contrast, replacements of glycine 11 and certain other internal residues result in loss of functions (D. M. Hampsey and F. Sherman, unpublished results). The inessential nature of the amino-terminal region, which is shown in Fig. 1, allows freedom for manipulating the end of the protein.

One example of a mutational pathway producing amino-terminal alterations is presented in Fig. 2. The *cycl*-13 mutant lacks iso-1-cytochrome *c* because of the destruction of the AUG initiator codon. Formation of AUG initiator codons at two abnormal sites resulted in the formation of a longer and a shorter protein, respectively. These and numerous other results established that translation initiates only at the most 5' AUG codon and that, in contrast to prokaryotes, there is no absolute requirement for a unique nucleotide sequence immediately adjacent to the AUG initiation codon.²⁰ The analysis of the proteins (Fig. 2) indicated that methionine is cleaved when the penultimate residue is alanine but not when it is isoleucine.

We have evaluated 30 different sequences in the amino-terminal region of iso-1-cytochromes *c* that were derived from revertants of initiator, nonsense and frameshift mutants and from recombinants having alterations in the amino-terminal region of iso-1-cytochrome *c*.¹⁸ These iso-1-cytochrome *c* sequences suggest that the cleavage of methionine is dependent solely on the penultimate amino-terminal residue. In the best example, methionine is cleaved from threonine termini in approximately 100 different iso-1-cytochromes *c*, including those having alterations at amino acid position 2, which is position 3 of the precursor, as shown in Table I. Two other examples, presented in Table I, demonstrate that methionine is not excised from any of the different

TABLE I. Cleavage or retention of the amino-terminal residues of methionine form the normal and altered forms of iso-1-cytochrome *c* having penultimate residues of threonine, asparagine or isoleucine. (Summarized from Tsunasawa *et al.*)¹⁶

Cleaved	Retained
Thr-Glu-Phe- Thr-Gln-Phe- Thr-Leu-Phe- Thr-Tyr-Phe- Thr-Ser-Phe- Thr-Pro-Phe- Thr-Asp-Ser- Thr-Asp-Ile- Thr-Gly-Ile- Thr-Ala-Lys-	Met-Asn-Glu-Lys- Met-Asn-Gln-Phe- Met-Asn-Lys-Phe- Ac-Met-Asn-Asn-Asn- Met-Ile-Thr-Glu- Met-Ile-Glu-Phe- Ac-Met-Ile-Arg-Ile- Ac-Met-Ile-Lys-Phe-

sequences having penultimate residues of asparagine or isoleucine. The analysis of different forms of iso-1-cytochrome *c*, summarized in Table II, indicate that methionine is cleaved from penultimate residues of glycine, alanine, valine and threonine but not from isoleucine, leucine, asparagine, lysine, glutamic acid, methionine or arginine. The study also revealed that certain mutationally altered iso-1-cytochromes *c* were acetylated, including the following sequences: Ac-Met-Ile-Arg-; Ac-Met-Ile-Lys-; Ac-Met-Met-Asn; and Ac-Met-Asn-Asn-. Apparently yeast contains acetyltransferases that acetylate these mutant forms of iso-1-cytochrome *c* because their amino-terminal regions resemble the amino-terminal regions of naturally occurring proteins which are normally acetylated.¹⁶

Amino Termini of Proteins

The pattern of methionine excision deduced from the altered iso-1-cytochromes *c* is consistent with the pattern deduced from the amino termini of other proteins from all other sources. However, we wish to stress that the specificity of methionine aminopeptidases cannot be inferred simply from the amino termini of any mature protein because of the possibility of other pro-

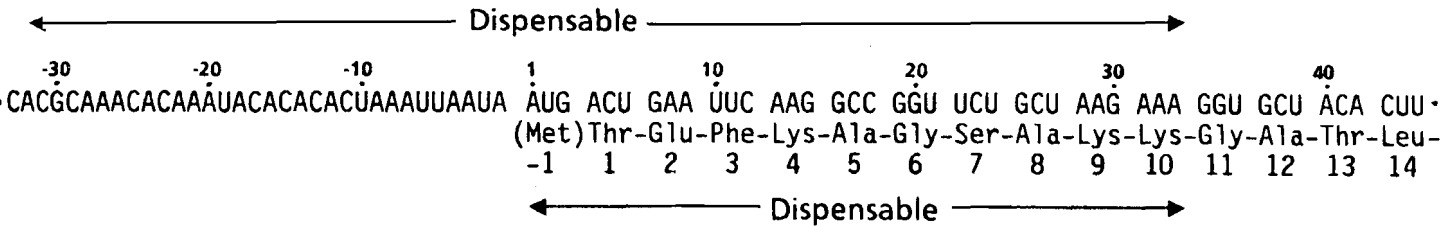


Fig. 1. The nucleotide sequence of a portion of the *CYCI* mRNA and the corresponding amino acid sequence of iso-1-cytochrome *c*. The analysis of mutations, including deletions and replacements, demonstrated that the region encompassing at least nucleotides -32 to +3 is not required for transcription or translation (S. B. Baim and F. Sherman, unpublished results) and that the amino-terminal region encompassing residues 1 to 10 is not required for protein function^{18,19} (D. M. Hampsey and F. Sherman, unpublished results). Because these regions are dispensable, numerous functional mutants with altered amino termini have been isolated.

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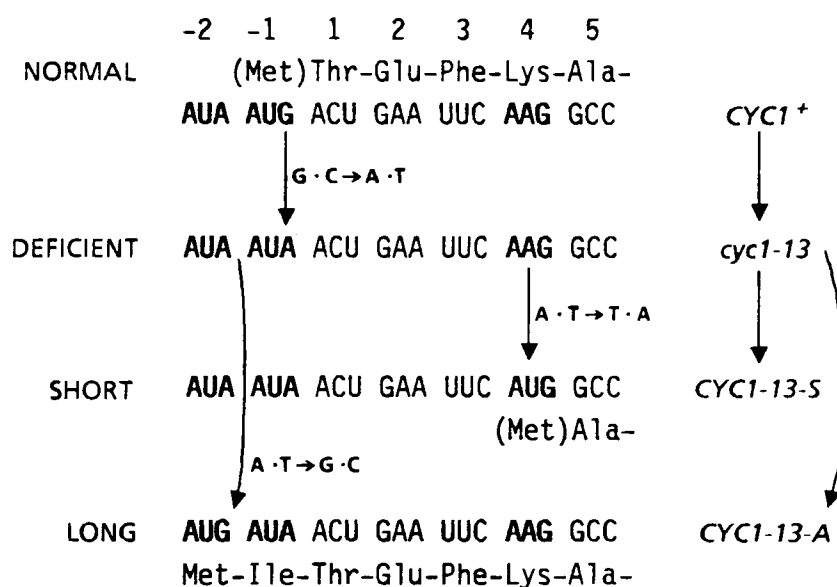


Fig. 2. An example of a pathway of mutations leading to iso-1-cytochromes c with altered amino termini. The cyc1-13 mutant lacks iso-1-cytochrome c because the G·C → A·T mutation destroyed the AUG codon required for initiation of translation. Revertants containing shorter and longer forms of iso-1-cytochrome c were generated by, respectively, A·T → T·A and A·T → G·C mutation that formed new AUG initiator codons at abnormal sites. The analysis of these iso-1-cytochromes c indicated that methionine is cleaved from penultimate residues of threonine and alanine but not isoleucine.¹⁷ Examples of other mutational pathways leading to other abnormal termini are presented by Tsunasawa et al.¹⁶

cessing events. Meaningful inferences can be made with proteins known to retain their termini, such as those uncovered by sequencing both the mature proteins and the corresponding mRNA or the corresponding exon region of DNA, or those uncovered by sequencing both the mature protein and the corresponding protein translated in a cell-free system lacking processing activities. Recently we have reported pertinent amino-terminal sequences of 82 different prokaryotic and eukaryotic proteins from a systematic search of the literature.¹⁶ The results obtained from the literature search are summarized in Table II in terms of the penultimate amino-terminal residue. For example, methionine is cleaved from all of the 17 different proteins having penultimate residues of alanine but not from any of five different proteins having penultimate residues of lysine.

Specificity of Methionine Aminopeptidases

These data from the literature and from the altered iso-1-cytochromes c suggest that methionine aminopeptidases from a wide range of prokaryotes and eukaryotes have a universal specificity for cleaving amino-terminal methionine from residues of glycine, alanine, serine, cysteine, threonine, proline and valine, but not from residues of aspartic acid, asparagine, leucine, isoleucine, glutamine, glutamic acid, methionine, lysine and arginine. Thus, except for several

proteins discussed below, the excision of methionine appears to be dependent primarily on the adjacent amino acid; this excision can be simply related to the sizes of the side chains of the penultimate residue as shown in Table II. In most cases, methionine can be cleaved from residues with a side chain having a radius of gyration of 1.29 Å or less, whereas it cannot be cleaved from residues with a side chain having a

radius of gyration of 1.43 Å or more. Thus, it is reasonable to suggest that large side chains of penultimate residues simply block the action of methionine aminopeptidase.

Exceptions to the Pattern of Methionine Cleavage

There are several apparent exceptions, listed in Table II, that do not follow the proposed rule for cleavage and retention of methionine. These exceptions appear to be due either to artifacts resulting from overproduction of the proteins or to properties of the specific proteins, but not properties confined to exceptional organisms.

Studies have directly demonstrated that methionine is partially retained on proteins that are artificially overproduced in *Escherichia coli*, probably due to the saturation of methionine aminopeptidase and deformylase capacities. When the synthetic α -consensus interferon is produced at high levels in *E. coli*, up to 70% of the molecules retain methionine on the Met-Cys-Asp-terminus, whereas when the same protein is produced at lower levels, amino-terminal methionine could not be detected (P. Lai, L. Goldstein, D. Fenton and N. K. Alton, unpublished results). In other studies of overproduced α - and β -subunits of tryptophan synthetase on a multicopy plasmid, the majority of these protein species were recovered with formylated methionine at the

TABLE II. Cleavage or retention of amino-terminal residues of methionine from proteins having different penultimate residues and the sizes (radii of gyration) of their side chains

Penultimate residue	Radius of gyration (Å) ^a	Iso-1-cytochrome c ^b		Other proteins ^b	
		Cleaved	Retained	Cleaved	Retained
Glycine	0	1	0	7	0
Alanine	0.77	1	0	17	0
Serine	1.08	—	—	13	1 ^c
Cysteine	1.22	—	—	2 ^d	0
Threonine	1.24	~ 100	0	4 ^e	1
Proline	1.25	—	—	3	2
Valine	1.29	1 ^f	0	6	0
Aspartic acid	1.43	—	—	3 ^g	6
Asparagine	1.45	0	4	0	2
Leucine	1.54	0	1	0	1
Isoleucine	1.56	0	4	—	—
Glutamine	1.75	0	1	0	3
Glutamic acid	1.77	—	—	0	4
Methionine	1.80	0	3	0	1
Lysine	2.08	0	2	0	5
Arginine	2.38	0	1	0	3

^a From Levitt.²¹

^b From Tsunasawa et al.¹⁶

^c Overproduced (see the text).

^d Synthetic interferon (P. Lai, L. Goldstein, D. Fenton and N. K. Alton, unpublished results) and an intermediate of actin from *Drosophila* (see the text).²²

^e One instance of 30% cleaved and 70% retained.

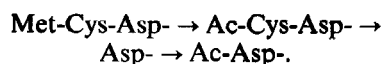
^f Consisting of 85% cleaved and 15% retained.¹⁷

^g Actin from mouse, chicken and *Dictyostelium*.

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amino termini.^{23, 24} In the overproduced condition, 60% and 40% of the β -subunit contained, respectively, F-Met-Thr-Thr- and Thr-Thr- termini, whereas the Thr-Thr- terminus is exclusively observed in the normal condition. Because the unformylated Met-Thr-Thr- sequence was not observed on the overproduced β -subunit, the deformylase, but not the methionine aminopeptidase, appeared to be limited in this particular condition. Another apparent inconsistency in the pattern of methionine cleavage can also be explained as an artifact of overproduction; the protein γ -glutamyl kinase from *E. coli* contains a Met-Ser-terminus,²⁶ whereas four other proteins from *E. coli* and nine proteins from various other organisms contain Ser- or Ac-Ser- termini.¹⁶ However, in order to facilitate the production of γ -glutamyl kinase for purification, the protein was overproduced in an expression vector.²⁶ Thus the termini of artificially overproduced proteins may not correspond to the termini of native proteins produced at the normal lower level.

Another apparent exception is the cleavage of methionine from several proteins with penultimate residues of aspartic acid (Table II). However, all of these exceptional proteins are actins whose amino termini are processed through a pathway that appears to involve another aminopeptidase. In contrast to the general pathway in which methionine cleavage occurs on the nascent polypeptide chains after only the first 30 to 40 amino acids are polymerized, the cleavage of methionine from a 'class I' mouse actin occurs only after synthesis of the complete polypeptide chain having a Ac-Met-Asp- terminus; the precursor is subsequently processed to yield an Ac-Asp- terminus.²² Also unique to certain actins is the removal of two amino-terminal residues. The amino terminus of a 'class II' actin from *Drosophila* is processed through the following pathway:²²



Apparently, the methionine residue is cleaved from the penultimate residue of cysteine by the general methionine aminopeptidase early in translation while the nascent chain is still associated with polyribosomes; after complete synthesis of the actin polypeptide the Ac-Cys- terminus is removed, probably by the same enzyme that removes Ac-Met- termini from 'class I' actins. These and other results suggest the occurrence of an aminopeptidase that

cleaves acetylated residues (Ac-Met-, Ac-Cys-, Ac-Gly- and Ac-Ala-) from termini of fully translated intermediates of actins.²⁶

Another inconsistency involves the Met-Thr- termini of *E. coli* proteins. Methionine is retained on the regulatory subunit of aspartate transcarbamoylase, and is approximately 70% retained on cystathionine γ -synthetase and is cleaved from β -galactosidase.¹⁶ Also, there are inconsistencies in the pattern of methionine cleavage from proteins with penultimate residues of proline. Rabbit aldolase, chicken histone H2B, and human histone H2B contain Pro-termini whereas chicken aldolase contains an Ac-Met-Pro- terminus and bacteriophage fd coat protein D contains a Met-Pro- terminus.¹⁶ In addition, methionine is cleaved from only approximately 85% of the Met-Val- termini of an altered iso-1-cytochrome *c*.¹⁷ These results suggest that protein structures other than the residue adjacent to the terminal methionine may interfere with the excision process. Because inconsistencies and partial cleavage of methionine residues so far have been observed only with proteins having penultimate residues of threonine, proline and valine (Table II), it is possible that the larger sizes of their side chains (1.24–1.29 Å) make the action of methionine aminopeptidase unusually sensitive to other regions of the proteins.

Conclusion and Future Work

The findings summarized in this review indicate that methionine aminopeptidases from diverse organisms all have the same specificity and that its action is restricted by large side chains of the adjacent residue. We have considered only the simplest geometric parameter of the side chain, the radius of gyration, which reflects its size and conformational

flexibility.²¹ Because the inhibition of cleavage can be reasonably predicted by size alone, as shown in Table II, there is no reason to involve other parameters, although other properties, such as the charge of the side group, may actually play a role in influencing the activity of methionine aminopeptidase. Clearly, the residue at amino acid position 3 of the precursor form does not affect methionine excision; methionine is cleaved from penultimate residues of threonine, even when position 3 is occupied by large residues (Table I). However, we have arbitrarily explained away a few inconsistencies by assuming that other regions of the protein may completely or partially inhibit excision of the amino-terminal residue of methionine, and that proteins with penultimate residues having intermediate sizes of side chains (threonine, proline and valine) may be more sensitive to other influences. In summary, we propose that methionine is never cleaved by the general methionine aminopeptidase when the size of the side chain of the penultimate residue is 1.43 Å or larger, whereas cleavage always takes place when the side chain is 1.22 Å or smaller; cleavage takes place when the side chains are 1.24–1.29 Å unless other properties of the protein inhibit its action. These relationships are clearly revealed in Fig. 3.

In our previous study¹⁶ we analyzed mutant forms of iso-1-cytochrome *c* having numerous alterations in the amino-terminal region of the protein. Site-directed mutagenesis is currently being used to generate a series of iso-1-cytochrome *c* with alterations solely at the penultimate residue (R. Moerschell and F. Sherman, unpublished results). These iso-1-cytochromes *c* will not only provide information on the retention and cleavage of methionine from the residues not included in Table

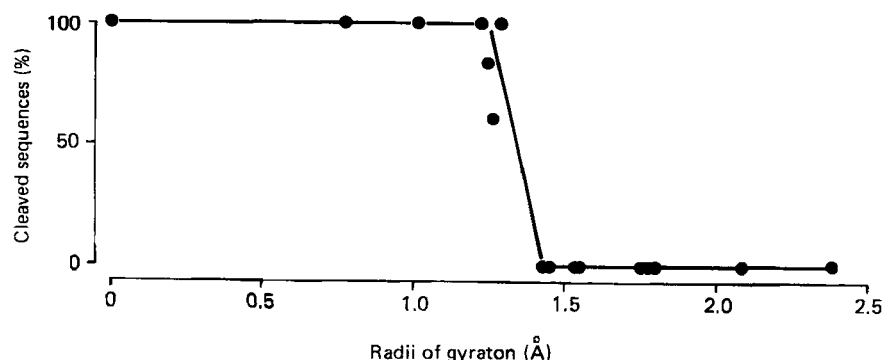


Fig. 3. The results from Table II, relating the percentage of sequences from which the amino-terminal residues of methionine are cleaved and the sizes of the side chain (radii of gyration) of the penultimate residues. The actin and overproduced proteins are excluded and the iso-1-cytochromes *c* with penultimate residues of threonine are represented as a single sequence.

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II, but will also provide a series of comparable proteins with only single replacements. Because Val- and Met-Val-termini were recovered, respectively, at 85% and 15%,¹⁷ it would be of interest to determine quantitatively the extent of methionine cleavage from such a series and to determine if the sizes of the side chains are related to the degree of partial cleavage.

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